

Phone-level accentedness scoring

From speech and expected phonemes to an interpretable

0-100 score for every sound.

AUDIO + PHONES

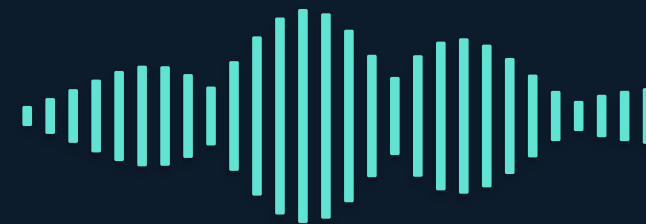
ORDINAL MODEL

GRADIO DEMO

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Modeling • evaluation • product demo

ILLUSTRATIVE OUTPUT



hello

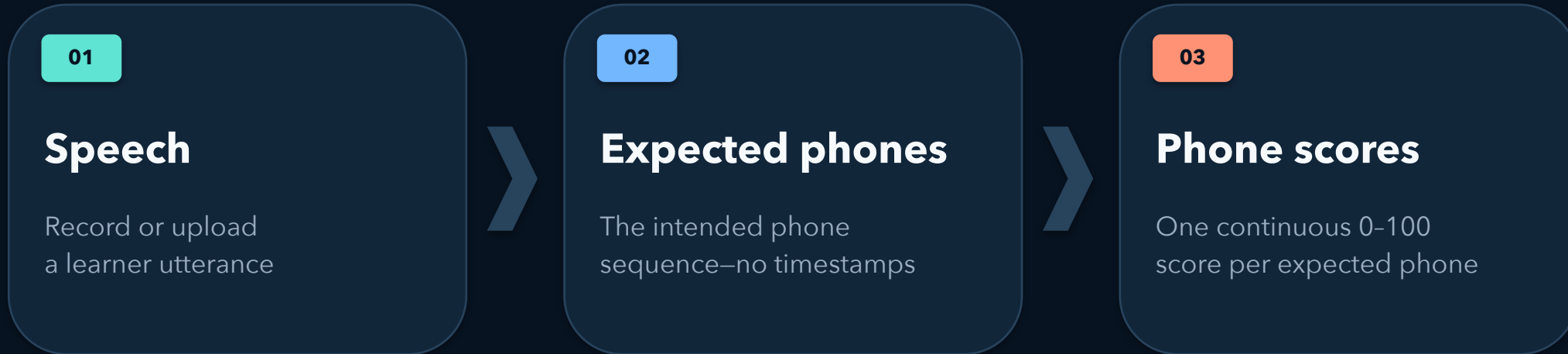
expected phones

h	ə	l	ou
92	45	89	12

$$\text{Score} = 50 \times [P(Y \geq 1) + P(Y \geq 2)]$$

Source: challenge brief example; scores shown are illustrative.

The task: localize pronunciation feedback



What makes it difficult

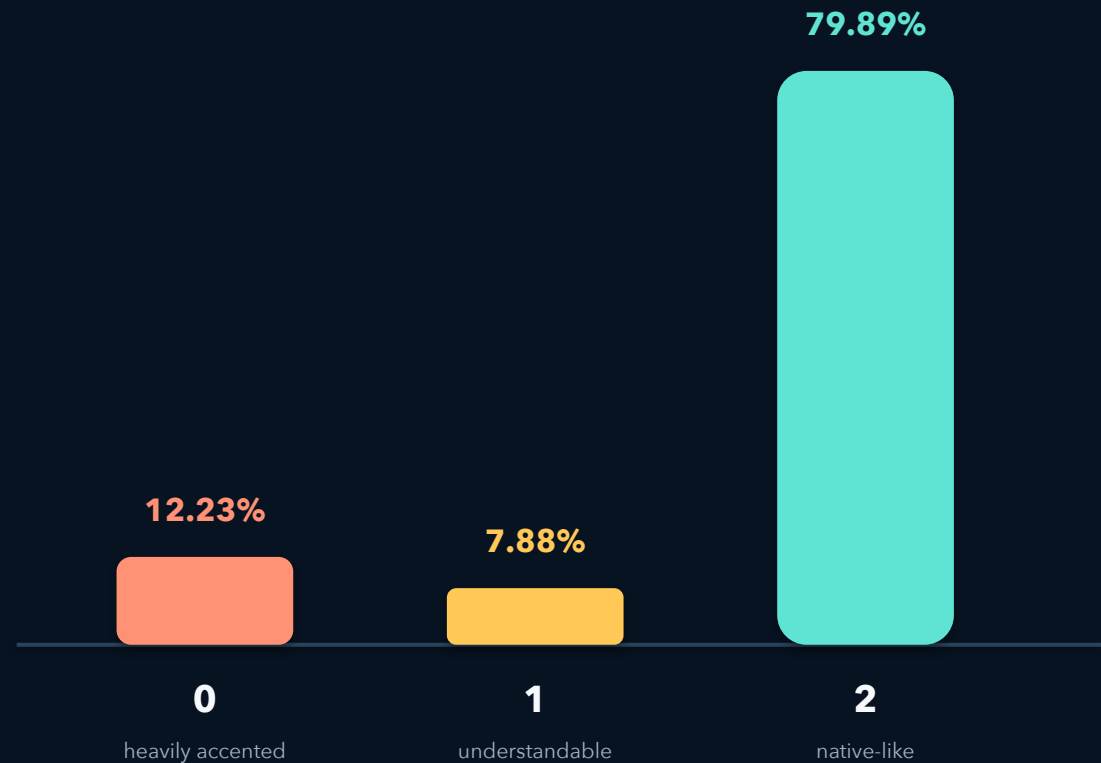
- Labels are ordered ($0 < 1 < 2$), but output must be continuous.
- Phone boundaries are absent and must be inferred from audio.
- The majority label dominates, so raw accuracy is misleading.

Product question

Where did this realization sound less American-like—and what should the learner practice next?

The data is small—and heavily imbalanced

TRAIN LABEL DISTRIBUTION



- NO PHONE TIMESTAMPS
- NO SPEAKER IDs
- NO RATER AGREEMENT

2,799

TRAIN UTTERANCES

100

VALIDATION UTTERANCES

90,239

LABELED PHONES TOTAL

44

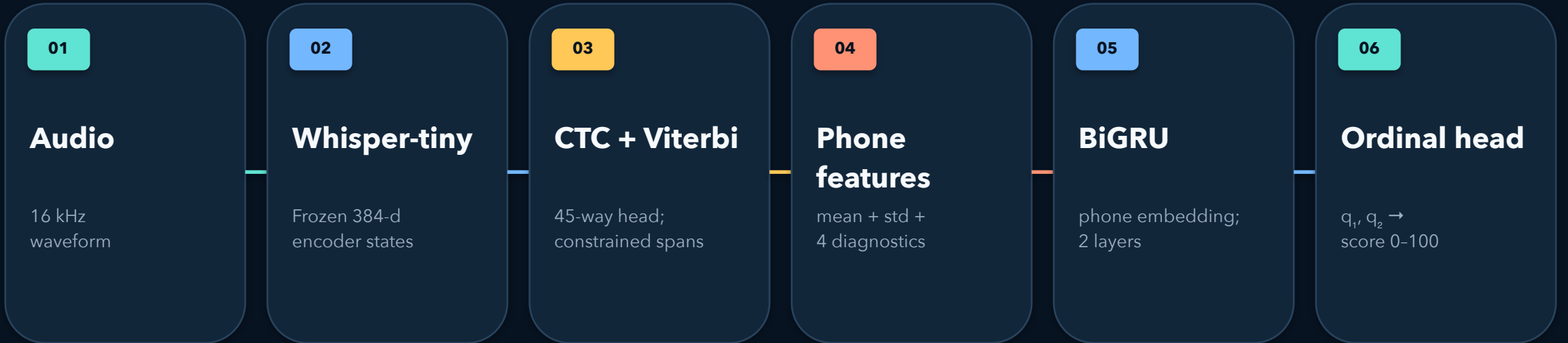
PHONE TYPES

79.47% validation accuracy by always predicting "native-like"

Yet recall for labels 0 and 1 is zero.

These gaps shape both modeling and evaluation.

Architecture: align first, then score

**9**

CTC epochs

18

scorer epochs

0

alignment fallbacks

42

random seed

Training discipline: select the objective with prompt-purged grouped OOF, then run one fixed fit on all 2,799 training utterances.

The objective and metrics match the ordering

ORDINAL REGRESSION

Two ordered questions

$$q_1 = P(Y \geq 1) \quad q_2 = P(Y \geq 2)$$



$$\text{score} = 50 \times (q_1 + q_2)$$

EVALUATION

Balanced MAE

Primary metric

Compute MAE within each true label, then average the three class errors equally.

- Preserves continuous error magnitude
- Stops the 80% majority class dominating

Secondary checks

QWK

MACRO-F1

BALANCED ACC.

SPEARMAN

Loss weighting: $n_c^{-0.54}$ per token • mean-one normalized

Validation: E16 improves balanced error

BALANCED MAE • LOWER IS BETTER

Strongest static



32.41

18.01

OVERALL MAE

0.579

QUADRATIC-WEIGHTED
KAPPA

Sequence only



32.03

0.568

MACRO-F1

0.558

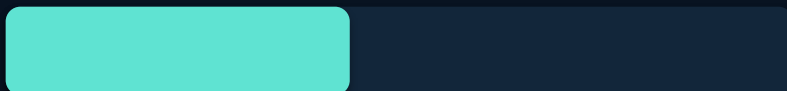
SPEARMAN

E01 incumbent



22.57

E16 selected



21.85

2,996 validation phones

10,000-draw utterance-bootstrap CI: 20.73-23.05

-0.725

balanced-MAE points vs previous E01
checkpoint

95% CI [-1.355, -0.081]

Important: this split is overlap-heavy—not new-speaker evidence.

What I tried—and what I selected

REJECTED

Bigger encoder

25.60 vs E01 22.57

Whisper-small worsened balanced MAE; no small-specific hyperparameter retuning was run.

REJECTED

Auxiliary labels

+0.039 BM delta

Severity and cluster targets produced no reliable gain; class-0 MAE worsened.

REJECTED

Full inverse weights

BM -2.02 • MAE +3.86

Rare-class recall improved, but QWK, macro-F1, majority recall, and calibration regressed.

SELECTED

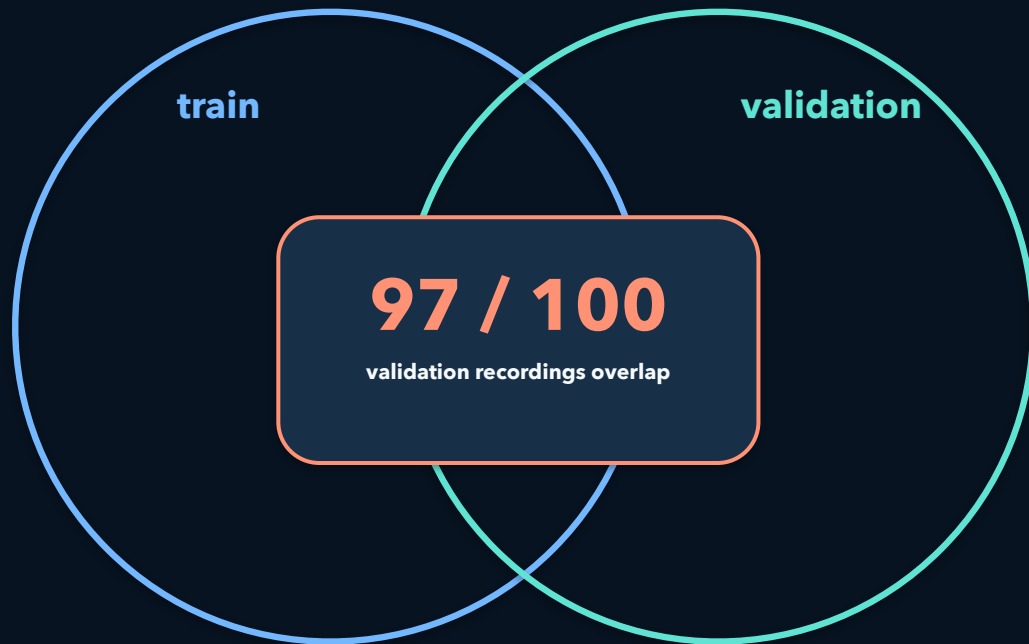
E16: Frozen Whisper-tiny + constrained CTC + alpha=0.54 ordinal head

Prompt-purged OOF and the one-shot final comparison passed every gate.

Each result comes from its documented protocol; this is not a single shared leaderboard.

The largest risk is the validation split

PSEUDO-SPEAKER OVERLAP



98.0% of validation phones

sit inside those inferred voice clusters

What this changes

- 92 / 100 validation prompts also appear in training.
- Reported metrics describe a partly seen-speaker / seen-prompt distribution.
- A speaker-disjoint benchmark is required before making generalization claims.

WavLM clusters are inferred voices—not verified identities.

Sniff test: right direction, weak separation

LEGACY E01 OWN-VOICE PAIR

+3.13
mean points

Best American



70.05

Non-native



66.92

- Only 10 / 20 phones moved higher
- Pace differed: 2.76s vs 4.08s

HEAVILY ACCENTED PHONES

46.77%

E16 SCORED BELOW 25

10.45%

E16 INCORRECTLY ≥ 75

Failure pattern

Historical E01 examples reacted when accent also hurt expected-phone recognition, but missed subtler non-American realizations that Whisper still recognized confidently.

E16 raises label-0 recall, but still misses subtle errors.

Automated judges did not validate the labels

GOPT teacher

5,863 / 5,894 → label 2

Useful ranking signal, but cross-corpus calibration collapsed near native-like; macro-F1 0.299.

NOT SAFE TO RELABEL

Local audio LLM

347 / 347 → label 2

Gemma 4 12B passed structure checks but failed the informativeness gate completely.

NOT SAFE TO RELABEL

GPT audio judge

40% agreement • 0 label-0

Balanced 30-item audit: macro-F1 0.299, QWK 0.143; it remained too lenient.

NOT SAFE TO RELABEL

Next valid step

Blinded expert review. The deterministic packets exist; no human-rating ledger exists yet.

Demo: turn a score into a practice loop

Pronunciation practice

Sentence to practice

We are both children together.

Generated phones • editable

w

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b

ou

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tʃ

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l

Record / upload audio

Score pronunciation

Score bands

practice

developing

strong

Product choices

- Microphone or upload
- Sentence playback
- Automatic, editable phonemes
- Ordered per-phone feedback

Difficulty ≠ model score

Keep raw scores fixed; adapt only the coaching threshold per phone and learner.

Run locally: `cd submission • uv run python demo_app.py`

Takeaway: useful signal, honest uncertainty

The model learned **useful acoustic signal**.

The evaluation has not yet proven **new-speaker robustness**.

01

Benchmark

Collect a genuinely speaker-disjoint, expert-rated evaluation set.

02

Calibrate

Review label disagreements and fit phone-specific calibration with uncertainty.

03

Improve

Test a more accent-sensitive encoder across seeds and matched-pace voice pairs.

Current scope A local coaching prototype with visible uncertainty—not a high-stakes judgment of identity, ability, or proficiency.

[Questions](#) • [discussion](#) • [live demo](#)